- A plane wavefront undergoes diffraction in Fraunhofer diffraction.
- The equation for nth minimum in single slit diffraction is, $a \sin\theta = n\lambda$ (n^{th} minimum)
- The equation for nth maximum in single slit diffraction is, $a\sin\theta = (2n+1)\frac{\lambda}{2}$ (n^{th} maximum)
- Fresnel's distance is the distance upto which ray optics is obeyed and beyond which ray optics is not obeyed but, wave optics becomes significant. $z = \frac{a^2}{2\lambda}$
- Diffraction can also happen in grating which has multiple slits of thickness comparable to wavelength of light used. $(a + b)\ \sin\theta = m\ \lambda$; $\sin\theta = N m \lambda$
- Wavelength of monochromatic light and also different colours of polychromatic light can be determined. Using diffraction grating and spectrometer.
- According to *Rayleigh's criterion*, for two point objects to be just resolved, the minimum distance between their diffraction images must be in such a way that the central maximum of one coincides with the first minimum of the other and vice versa
- Angular resolution, $\theta = \frac{1.22\lambda}{a}$
- Special resolution, $r_0 = \frac{1.22\lambda f}{a}$
- Resolution is measured by the smallest distance which could be seen clearly without
- the blur due to diffraction.
- The phenomenon of restricting the vibrations of light (electric or magnetic field vector) to a particular direction perpendicular to the direction of propagation of wave is called *polarization* of light.
- If the vibrations of a wave are present in only one direction in a plane perpendicular to the direction of propagation of wave is said to be *polarised* or *plane polarised light*.
- The plane containing the vibrations of the electric field vector is known as the *plane of vibration*.
- The plane perpendicular to the plane of vibration and containing the ray of light is known as the *plane of polarisation*.
- In *plane polarised* light the intensity varies from maximum to zero for every rotation of 90° of the analyser.
- In *partially polarised* light the intensity varies from maximum to minimum for every rotation of 90° of the analyser.
- The intensity of transmitted light through two cross polaroids is given by Malus' Law. $I = I_0 \cos^2\theta$
- Nicol prism separates ordinary and extraordinary rays by double refraction.
- Light scattered by molecules at perpendicular direction to the incident light is found to be plane polarised.

XII_Physics_Vol-2_UNIT-7_EM.indd 99 03-03-2022 15:19:39